MASTERY — Quantitative Analyst (Fund Intelligence & Optimization)

On-the-Job Training & Operational Performance Playbook

Company: PortfolioFuture | Scope: 10,000+ Funds, 130M+ Relationships, 290M+ Opportunities
Core Architecture: Fund Substitution, Return Decomposition, SEC 13F/ADV Analytics & Portfolio Optimization
Candidate: Bhupesh Chandra | Chief Learning Officer Mastery Program

A. JOB AT A GLANCE

Mission: PortfolioFuture is an independent fund discovery and intelligence company evaluating 10,000+ ETFs/mutual funds across 130M+ fund relationships and 290M+ ownership opportunities. Connecting with 20,000+ RIAs and 400,000+ advisors, the role applies quantitative finance, machine learning, econometric decomposition, and large-scale portfolio optimization to determine where funds can credibly compete for allocations.

Category	Specification & Technical Mandate
Core Domains	Fund Substitution, Return Decomposition (Factor Alpha), SEC 13F/ADV Ownership Intelligence, Large-Scale Optimization
Tech Stack	Python (NumPy, SciPy, Pandas, Polars, Statsmodels, CVXPY), PyTorch/Scikit-Learn, DuckDB/SQL, LLM Prompt Pipelines
Primary Datasets	CRSP/Morningstar Fund Daily Returns & Fees, SEC Form 13F Institutional Holdings, Form ADV/IAPD, N-PORT schedules
Research Philosophy	Strict Independence: Commercial relationships never bias substitution rankings, pure alpha extraction, or research conclusions

B. JOB INTELLIGENCE MAP & PRIORITY RANKING

Task / Area	Tier	Core Deliverable & Business Consequence	Frequency
1. Fund Substitution Modeling	**Tier 1**	Pairwise return similarity, tracking error, fee drag delta. Drives advisor replacement decisions.	Continuous / Daily
2. Investable Return Decomposition	**Tier 1**	Separates beta exposures from true residual alpha (R^2 regression, factor spanning).	Daily / Monthly Close
3. SEC 13F & Form ADV Ownership Analytics	**Tier 1**	Extracts $290M+ in ownership relationships to identify high-fee displacement targets.	Quarterly Filing Surge
4. Large-Scale Portfolio Optimization	Tier 2	Quadratic/CVaR/HRP optimization testing marginal Sharpe & tracking error budgets.	Weekly / Model Cadence
5. LLM Research & Quantum Optimization	Tier 2	Unstructured filing parsing (N-PORT) and combinatorial QAOA/QUBO exploration.	Research Sprints

D. END-TO-END WORKFLOW: FUND DECOMPOSITION & SUBSTITUTION

Step 1 (Data Ingest): Ingest daily Adjusted NAV returns for Target Fund F and benchmark factor universe (Market MKT, Size SMB, Value HML, Momentum MOM, Quality QMJ, and Liquid Passive ETF Proxies).

Step 2 (Constrained Multi-Factor Regression): Run rolling 36-month non-negative constrained regression: $R_{F,t} - R_{f,t} = ■lpha_F + \sum_{k=1}^K ■eta_k (R_{k,t} - R_{f,t}) + \epsilon_{F,t}$ subject to $\sum ■eta_k = 1$ and $■eta_k \ge 0$.

Step 3 (Residual Alpha Extraction): Isolate true managerial residual performance: $Pure\ ■lpha_F = ■lpha_F - \Delta Expense_Ratio$. Compute Information Ratio: $IR = rac{■lpha_F}{\sigma(\epsilon_F)}$.

Step 4 (Candidate Matching): Search universe of 10,000+ funds for Candidate Substitutes S where $Cosine\ Similarity(R_F, R_S) \ge 0.95$ and $Expense_Ratio(S) < Expense_Ratio(F)$.

Step 5 (13F Addressable Market Mapping): Match Target Fund F to SEC Form 13F/ADV database. Sum total AUM held across wealth managers paying excessive fees. Output total addressable switching market in USD.

Step 6 (Audit Validation): Stress-test out-of-sample persistence across 12-month forward holdout windows. Ensure results pass t-stat significance ($p < 0.05$).

I. ONE-PAGE ELITE JOB CHEAT SHEET

Metric / Rule	Mathematical Formulation & Operational Standard
Return Similarity	Cosine Similarity: $\frac{\mathbf{r}_A \cdot \mathbf{r}_B}{\|\mathbf{r}_A\| \|\mathbf{r}_B\|}$. Measures co-movement independent of scale; threshold ≥ 0.96 for close substitutes.
Residual Alpha (Pure α)	$\alpha_{residual} = R_F - \sum \beta_k F_k$. Must test whether α survives the fee hurdle ($Expense\ Ratio$).
Tracking Error (TE)	$TE = \sqrt{252} \times \text{StdDev}(R_A - R_B)$. Core substitute candidate must maintain $TE < 2.50\%$ annualized.
Addressable Market	$\sum_{i \in RIAs} AUM_{i, TargetFund}$. The dollar size of vulnerable high-fee capital that can be displaced.
13F Lag Rule	SEC 13F filings have a 45-day reporting lag. Must model holding decay and flag high-turnover funds.
Independence Wall	Commercial fee-paying relationships never touch ranking logic. Code must verify zero sponsor weight overrides.

J. 30 / 60 / 90-DAY PROGRESSIVE PERFORMANCE ROADMAP

Days 1–30 (Foundation & Pipeline Velocity): Master internal DuckDB/SQL data schemas, write automated ingestion pipelines for SEC 13F / Form ADV, reproduce baseline Fund Decomposition regression models with 100% precision.

Days 31–60 (Model Robustness & Substitution Engine): Enhance pairwise fund substitution algorithms with non-linear clustering (t-SNE/UMAP) and out-of-sample persistence testing; build automated 13F advisor displacement screens.

Days 61–90 (Innovation & Scale): Deploy LLM-driven prospectus/N-PORT fee extraction pipelines; experiment with CVXPY large-scale portfolio quadratic optimization; publish empirical research note on ETF factor replication.